

Jyothir Anurag Vasu Chinnaboina

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Education

University of Wisconsin–Madison – *B.S. in Computer Science and Data Science*

Madison, WI

Awards: Dean's List (4 semesters)

Expected May 2026

Coursework: AI/ML, Reinforcement Learning, Algorithms, Operating Systems

GPA: 3.7/4.0

Statistics, Database Management, Data Structures

Experience

Software Developer Intern (Part-time) – *Vijay Rekha Life Sciences*

Aug 2025 – Present

Sole Developer · vijayrekha.chinnaboina.com

- Deployed a neonatal screening lab service (FastAPI, PostgreSQL, React/TypeScript) processing 300+ samples daily with automated validation and abnormality flagging for 4 technicians.
- Automated 3+ hours of manual work into a single-click Python pipeline (pandas, xlsxwriter) that transforms raw mass spectrometry data for 40+ compounds into formatted Excel reports.
- Accelerated clinical report validation by 85% with a batch pipeline (pdfplumber, regex) that cross-validates patient data from 100+ PDF reports in a single run.

Mainframe Intern (Part-time) – *Wisconsin Department of Administration*

Jun 2025 – Present

- Reduced manual triage time by 90% with a Java 8 usage tracker (Python, z/OS) that parsed job logs into CSV and prioritized remediation follow-ups across 5+ teams.
- Eliminated 4+ hours/week of manual reporting via a Python orchestrator with YAML config, JSON Schema validation, Jinja2 templating, and automated email distribution.

Computer Science Tutor – *Undergraduate Learning Center, UW–Madison*

Sep 2024 – Aug 2025

- Tutored 600+ students across 8 CS courses and evaluated 160+ tutor candidates across 3 hiring cycles, designing technical interview questions for the Learning Center.

ERP Integration Intern – *United States Pharmacopeia*

Jun – Aug 2024

- Improved data retrieval time by 70% with an analytics dashboard (Python, pandas) using optimized SQL queries joining 5 tables to surface order hold trends.

Projects

F1 Race Strategist (RL) | *Qwen3-4B, GRPO, LoRA, TRL, OpenEnv, FastAPI, Docker*

Present

Meta PyTorch OpenEnv Hackathon – Long-Horizon Planning Track · f1.chinnaboina.com

- Trained Qwen3-4B to 0.618 (48% gain), matching expert on weather scenarios (0.965 vs 0.950) via a deterministic 6-D Python reward.
- Engineered a two-stage fine-tuning pipeline (SFT on ~5K trajectories, then 800 GRPO steps via TRL/LoRA) peaking at reward 0.907 by step 480.
- Closed an episode-postmortem self-improvement loop classifying failures (missed undercut, late weather) into hints re-injected on next reset.

ApplyOps | *Python, Claude Agent SDK, ChromaDB, OpenAI Embeddings, asyncio, SQLite*

Present

Personal pipeline infrastructure · github.com/Anurax1321/ApplyOps

- Designed and built a multi-agent tailoring pipeline (Claude, ChromaDB) saving ~80K tokens/app (~70%) across 250+ apps via Writer/Critic loops.
- Spun up a Python socket daemon (5 vector collections, sub-second latency from 30s cold) auto-injecting bullets and memory into every Claude conversation.
- Validated 3 concurrent runs to zero SQLite errors via WAL + rebuild locks, covered by 26 pytest tests across retrieval, cache, and daemon fallback.

Office Hours Queue | *FastAPI, ChromaDB, OpenAI API, MCP, Docker, React*

Sep 2025 – Present

Co-Founder · *Advisor: Prof. Louis Oliphant (CS)* · ohq.cs.wisc.edu

- Built an agentic code-analysis pipeline (MCP, 9 tools) running an OpenCode LLM in sandboxed Docker, used across 2 CS courses with 500+ students.
- Built a RAG assistant (FastAPI, ChromaDB, Qwen3-Embedding-8B) with DBSCAN clustering for auto-FAQ generation across 4 sources, ~2s lookup.
- Architected a 6-service containerized backend (React, Flask, FastAPI, ChromaDB, Redis, Traefik) with SSE streaming and GitHub Actions CI/CD for one-command deploys.

Skills

Languages: Python, Java, JavaScript, TypeScript, Go, SQL, R, C, Bash, HTML/CSS

AI/ML: PyTorch, Hugging Face Transformers, TRL, GRPO, PEFT/LoRA, Accelerate, scikit-learn, ChromaDB, OpenAI API (GPT-4o), Claude Agent SDK, Multi-agent Orchestration, RAG, MCP, LLM Agents, Embeddings, pandas, NumPy, Hugging Face Hub/Spaces

Backend & Infra: FastAPI, Flask, Pydantic, Gradio, Node.js/Express, React, Docker, Redis, Traefik, GitHub Actions, Terraform, OpenEnv

Databases & Cloud: PostgreSQL, Redis, Oracle DB, MySQL, SQLite, Google Cloud Platform

Other: REST APIs, SSE, WebSockets, OAuth2, JWT, Pytest, Agile/Scrum, Linux